ModelIt!

Student Activity Pack

What Happens When Wolves Disappear?

Modeling Trophic Cascades, Population Dynamics, and Ecosystem Stability

Grade 10L1 | NGSS HS-LS2-1, HS-LS2-2 | Stability and Change

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

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Table of Contents

ModelIt!

1 Pre-Assessment 3

2 Scoring Rubric 4

3 Component Analysis 5

4 Model Recording Page 6

5 Simulation Observations 7

6 Research & Extend 8

7 Reflection Questions 9

8 STEM Challenge 10

1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 What do you think would happen to an ecosystem if its top predator was completely removed?

2 How can one species affect an entire landscape — including rivers and soil?

3 Draw a food web showing at least four trophic levels and predict what happens if you remove the top level.

4 What does 'balance of nature' mean, and do ecosystems naturally maintain balance without human interference?

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Population Size | Food Availability | Predator Pressure | Habitat Area | Extinction Risk

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Wolves Present

Predator Pressure: Natural wolf population

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Wolves Removed

Predator Pressure: Zero (wolves eliminated)

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Wolf Reintroduction

Predator Pressure restored after degradation

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Behavioral Response — Changes in prey behavior (where they graze, how long they stay in one area) caused by the presence or fear of predators — the 'ecology of fear' that may be more important than direct killing

Riparian Vegetation — Trees, shrubs, and grasses growing along riverbanks that depend on grazing pressure remaining below critical levels — willows and aspens are keystone species in Yellowstone's riparian zones

Species Diversity Count — The total number of species present in the ecosystem, which typically increases when trophic cascades maintain habitat diversity and decreases when overgrazing homogenizes the landscape

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

When Yellowstone removed wolves, the rivers moved. When wolves returned, the rivers came back. How can one species reshape an entire landscape? Use evidence from your simulation to support your answers.

1 How does your model demonstrate that ecosystems are complex interconnected systems rather than simple food chains?

2 Why might the 'ecology of fear' be as important as direct predation in maintaining ecosystem balance?

3 What does the Yellowstone case reveal about the unintended consequences of wildlife management decisions?

4 How does your model help explain why removing one species can cause rivers to change course?

5 What are the limitations of modeling a complex ecosystem with only five components?

8 STEM Challenge: Design a Wildlife Corridor and Reintroduction Plan

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YOUR ENGINEERING MISSION

Design a wildlife management plan for reintroducing a keystone predator to a degraded ecosystem, including corridor connections, population targets, and monitoring protocols.

THE SCENARIO

A state wildlife agency is considering reintroducing wolves to a region where they were eliminated 80 years ago. The ecosystem has degraded significantly — elk overgraze, riparian habitats are damaged, and biodiversity has declined. However, local ranchers and communities oppose reintroduction. Your team must design a science-based plan that maximizes ecological benefit while addressing human concerns.

Engineering Design Thinking

1. WHAT IS THE TARGET WOLF POPULATION AND HOW DID YOUR MODEL INFORM THIS NUMBER?

2. HOW WILL YOU ADDRESS CONFLICTS BETWEEN WOLVES AND LIVESTOCK OPERATIONS?

3. WHAT CORRIDOR CONNECTIONS ARE NEEDED AND WHAT MINIMUM WIDTH ENSURES VIABLE WOLF MOVEMENT?

4. WHAT ECOLOGICAL INDICATORS WILL YOU MONITOR AND AT WHAT INTERVALS TO ASSESS CASCADE EFFECTS?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Wildlife Ecologists study the interactions between species and their environments, managing ecosystems and designing conservation strategies. They work for federal and state wildlife agencies, conservation organizations, and research institutions, earning $50,000–$95,000/year. Conservation Biologists who specialize in endangered species earn $55,000–$100,000/year.